

FINAL REPOSITORY ARCHITECTURE DOCUMENT

Trustworthy AI for Protein Engineering

Discovering Mutation-Space Reliability and Applying It to Industrial Carbon-Capture Enzyme Design

Complete Repository Design, Directory Structure, Module Organization, Data Flow, And System Architecture Specification

Version 2.0

Purpose

This document defines the complete repository architecture for the project.

It serves as the authoritative reference for:

Repository Structure

Directory Organization

Module Organization

Data Organization

Scientific Workflow

Validation Workflow

Publication Workflow

Competition Workflow

Every file, dataset, module, report, figure, document, and generated artifact must have a defined location within this architecture.

The goal is to ensure:

Reproducibility

Maintainability

Scalability

Traceability

throughout the entire project lifecycle.

Architectural Philosophy

The repository is not organized as:

Code

+

Data

The repository is organized as:

Scientific Research System

The architecture separates:

Scientific Design

Software Implementation

Datasets

Validation

Analysis

Publication

Competition

Release

into clearly defined layers.

Complete Repository Structure

trustworthy_ai_protein_engineering/

```
|— configs/
|— data/
|— docs/
|— src/
|— tests/
|— outputs/
|— results/
|— reports/
|— figures/
|— papers/
|— science_fair/
|— presentations/
|— release/
|— checkpoints/
|— models/
|— notebooks/
|— logs/
|— scripts/
└— README.md
```

Architectural Layers

The repository is divided into ten major layers.

Documentation Layer

Configuration Layer

Source Code Layer

Data Layer

Analysis Layer

Validation Layer

Publication Layer

Competition Layer

Release Layer

Archive Layer

Documentation Layer

Location:

docs/

Purpose:

Scientific Documentation

Technical Documentation

Project Documentation

Documentation Structure

docs/

FINAL_PROJECT_BLUEPRINT.md

PROJECT_DOCUMENT_GUIDE.md

MASTER_IMPLEMENTATION_SPECIFICATION.md

SCIENTIFIC_ALGORITHMS_SPECIFICATION.md

DATA_ARCHITECTURE_AND_DATASET_SPECIFICATION.md

IMPLEMENTATION_ROADMAP.md

CURSOR_IMPLEMENTATION_PROMPTS.md

SYSTEM_OPERATION_TESTING_AND_VALIDATION_HANDBOOK.md

PROJECT_MASTER_INDEX.md

PROJECT_RELEASE_PACKAGE.md

PROJECT_EXECUTION_MANUAL.md

FINAL_REPOSITORY_ARCHITECTURE_DOCUMENT.md

Configuration Layer

Location:

configs/

Purpose:

Centralized Configuration

Structure

configs/

development.yaml

validation.yaml

production.yaml

paths.yaml

`datasets.yaml`

`figures.yaml`

`publication.yaml`

Source Code Layer

Location:

`src/`

Purpose:

Scientific Computation

Data Processing

Analysis

Visualization

Complete Source Tree

`src/`

`infrastructure/`

`validation/`

`datasets/`

`sparsity/`

`experiment1/`

`experiment2/`

`integrated/`

reliability/

tktk/

active_site/

industrial_fitness/

visualization/

publication/

release/

Infrastructure Module

Location:

src/infrastructure/

Purpose:

Shared System Services

Structure:

config_loader.py

logger.py

experiment_tracker.py

checkpoint_manager.py

dataset_registry.py

hardware_detection.py

Validation Module

Location:

`src/validation/`

Purpose:

Data Validation

Schema Validation

Result Validation

Certification

Structure:

`schema_validator.py`

`dataset_validator.py`

`figure_validator.py`

`result_validator.py`

`certification_manager.py`

Dataset Module

Location:

`src/datasets/`

Purpose:

Dataset Management

Data Loading

Data Transformation

Structure:

loaders/

transformations/

exports/

Sparsity Framework Module

Location:

src/sparsity/

Purpose:

Generate Sparsity Metrics

Structure:

src/sparsity/

 empirical/

 evolutionary/

 structural/

 combined/

Empirical Sparsity Submodule

src/sparsity/empirical/

 empirical_sparsity.py

 empirical_validation.py

Evolutionary Sparsity Submodule

`src/sparsity/evolutionary/`

`esm_probability_engine.py`

`evolutionary_sparsity.py`

`evolutionary_validation.py`

Structural Sparsity Submodule

`src/sparsity/structural/`

`sasa_engine.py`

`structural_sparsity.py`

`structural_validation.py`

Combined Sparsity Submodule

`src/sparsity/combined/`

`combine_sparsity.py`

`combined_validation.py`

Experiment 1 Module

Location:

`src/experiment1/`

Purpose:

Thermodynamic Inconsistency Analysis

Structure:

forward_predictions.py

reverse_predictions.py

antisymmetry_error.py

statistics.py

figures.py

Experiment 2 Module

Location:

src/experiment2/

Purpose:

Epistasis Reliability Analysis

Structure:

experimental_epistasis.py

predicted_epistasis.py

epistasis_error.py

statistics.py

figures.py

Integrated Analysis Module

Location:

`src/integrated/`

Purpose:

Reliability Discovery

Structure:

`build_dataset.py`

`integrated_statistics.py`

`pathway_analysis.py`

`integrated_figures.py`

Reliability Framework Module

Location:

`src/reliability/`

Purpose:

Reliability Estimation

Structure:

`reliability_score.py`

`reliability_classifier.py`

`reliability_validation.py`

TKT Module

Location:

src/tkt/

Purpose:

Transketolase Application

Structure:

mutation_enumerator.py

landscape_generator.py

landscape_analysis.py

candidate_generator.py

Active-Site Module

Location:

src/active_site/

Purpose:

Distance Analysis

Mechanistic Controls

Structure:

active_site_mapper.py

distance_engine.py

active_site_statistics.py

Industrial Fitness Module

Location:

src/industrial_fitness/

Purpose:

Engineering Candidate Ranking

Structure:

fitness_calculator.py

fitness_validator.py

rank_candidates.py

Visualization Layer

Location:

src/visualization/

Purpose:

Publication Graphics

Science Fair Graphics

Presentation Graphics

Structure

src/visualization/

 publication_figures/

 science_fair_figures/

 presentation_figures/

 killer_figure/

Publication Layer

Location:

`src/publication/`

Purpose:

Manuscript Support

Figure Export

Statistics Export

Release Layer

Location:

`src/release/`

Purpose:

Archive Creation

Release Packaging

Version Export

Data Layer

Location:

`data/`

Purpose:

Raw Data

Intermediate Data

Processed Data

Data Structure

data/

raw/

external/

intermediate/

processed/

final/

Raw Data

data/raw/

megascale_d/

sequences/

structures/

epistasis/

Intermediate Data

data/intermediate/

empirical/

evolutionary/

structural/

combined/

Final Data

data/final/

reliability/

tk/

industrial_fitness/

Results Layer

Location:

results/

Purpose:

Scientific Outputs

Structure:

results/

empirical_sparsity/

evolutionary_sparsity/

structural_sparsity/

combined_sparsity/

experiment1/

experiment2/

integrated_analysis/

reliability/

tkr/

active_site/

industrial_fitness/

Reports Layer

Location:

reports/

Purpose:

Certifications

Validation Reports

Audits

Structure:

reports/

validation/

certification/

audits/

reproducibility/

competition/

Figures Layer

Location:

figures/

Purpose:

All Visual Outputs

Structure:

figures/

 publication/

 science_fair/

 presentation/

 killer/

Publication Layer

Location:

papers/

Purpose:

Journal Materials

Structure:

papers/

 manuscript/

supplementary/

journal_submission/

Science Fair Layer

Location:

science_fair/

Purpose:

Competition Assets

Structure:

science_fair/

abstract/

board/

binder/

handouts/

judge_materials/

Presentation Layer

Location:

presentations/

Purpose:

Talks

Defense Materials

Competition Presentations

Structure:

presentations/

one_minute/

four_minute/

ten_minute/

defense/

Release Layer

Location:

release/

Purpose:

Final Project Distribution

Structure:

release/

datasets/

source/

documents/

figures/

archives/

Checkpoint Layer

Location:

checkpoints/

Purpose:

Recovery

Restart

Long Execution Support

Model Layer

Location:

models/

Purpose:

External AI Models

Cached Models

ESM Resources

Notebook Layer

Location:

notebooks/

Purpose:

Exploration

Prototyping

Interactive Analysis

Log Layer

Location:

logs/

Purpose:

Execution Tracking

Structure:

logs/

system/

experiments/

validation/

End-To-End Scientific Workflow

The official workflow of the repository is:

Raw Data

↓

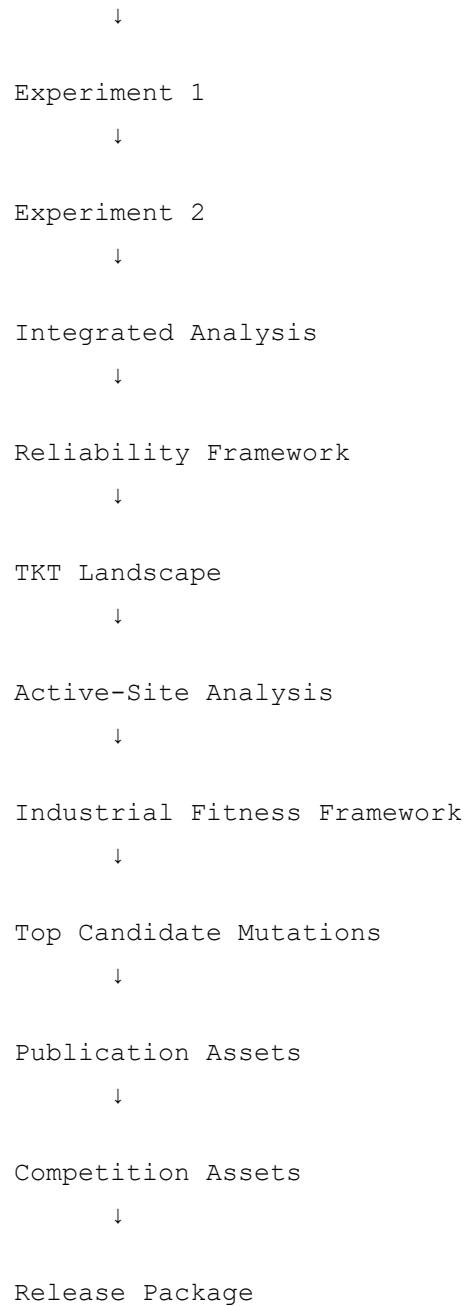
Empirical Sparsity

Evolutionary Sparsity

Structural Sparsity

↓

Combined Sparsity



Repository Certification Requirements

The repository architecture is certified only if:

Directory Structure Defined



Module Boundaries Defined



Data Flow Defined



Output Locations Defined



Validation Locations Defined



Publication Locations Defined



Competition Locations Defined



Release Locations Defined



Final Architectural Statement

This repository is designed as a complete scientific research platform for investigating:

Mutation-Space Sparsity



Prediction Failure



Reliability Estimation



Reliability-Guided
Protein Engineering

with a practical application in:

Industrial Carbon-Capture
Enzyme Design

using Transketolase as the primary engineering demonstration system.

END OF FINAL REPOSITORY ARCHITECTURE DOCUMENT

Version 2.0

Approved Repository Architecture For The Complete Trustworthy AI For Protein Engineering Project.